CHELSEA OGBONNA

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EDUCATION

Northwestern University **Evanston, IL**
B.S. Biomedical Engineering & M.S. Mechanical Engineering *Anticipated Graduation: June 2026*
Graduate GPA: 3.9

- **Relevant Courses:** Advanced Mechatronics, Biomedical Signals and Systems, Thermodynamics

PROFESSIONAL EXPERIENCE

Microfluidic Device Design for Syncytial Knot Isolation, *Team Leader* *Sept 2025 – Jun 2026*

- Conceptualized, designed, and led fabrication of three passive microfluidic devices through multiple design iterations to isolate 50–150 µm syncytial knots from placental explant tissue for maternal-fetal health research.
- Refined CAD geometries via COMSOL simulations over 10 design cycles and 18 experimental trials.

Rogers Lab, *Research Assistant* *Jun 2025 – Jun 2026*

- Fabricated and benchtop-validated 12 wireless TAVI-integrated pacemaker prototypes through 3D printing, PCB soldering, and flow testing, achieving 80% functional yield to address post-TAVR conduction injuries.

Solutions Lab, *Research Assistant* *Jun 2025 – Jun 2026*

- Trained 150+ Evanston Police Department officers in de-escalation techniques to improve conflict-resolution.
- Coded and analyzed 200+ hours of body-worn footage to quantify training impact, finding a 32% change in trust and rapport from community members post-training.

Project Commission on Urgent Relief and Equipment (C.U.R.E), *Human Factors Intern* *Jun 2025 – Sept 2025*

- Inspected, repaired, and certified 150+ biomedical devices for deployment to hospitals across 10+ countries.
- Diagnosed and resolved equipment failures, restoring an estimated $2M in medical equipment value.

Distal Biceps Tendon Repair, *Research Assistant* *Jun 2024 – Aug 2024*

- Executed a systematic literature review across 50–70 peer-reviewed studies to extract, clean, and analyze complex clinical datasets evaluating distal biceps tendon repair techniques and their surgical outcomes.

PROJECTS

Everyday Moving Helper *Mar 2026 – Jun 2026*

- Prototyped a modular, powered cargo platform with an expandable frame to improve transport of groceries, luggage, packages, and household items.
- Conducted load and terrain analyses validating safe transport of loads up to 30 lb across all surface types.

Haptic Diabetic Foot Ulcers *Jan 2026 – Mar 2026*

- Designed a wearable gait-monitoring system using an MPU6050 IMU and force-sensitive resistors to capture biomechanical walking data for early detection of diabetic foot complications.
- Developed Python analysis code to evaluate gait metrics, detecting deviations of 20% associated with abnormal walking patterns.

Biosignal Acquisition and Control Systems: EMG, EKG, and Blood Pressure *Jan 2025 – Jun 2025*

- Designed, built, and validated analog EMG, EKG, and blood pressure acquisition systems, achieving >90% cost savings versus commercial equivalents.
- Developed a Python FFT pipeline using the nLab API, reducing biosignal analysis time by ~80%.

Design Thinking and Communication 1, *Team Leader* *Jan 2023 – Mar 2023*

- Led redesign of a condominium recycling center with custom bins, signage, and a cardboard collection system using SolidWorks and Siemens NX, boosting recycling participation by 40% and winning Best Design Team.

LEADERSHIP

National Society of Black Engineers (NSBE), *Academic Excellence Chair* *Apr 2024 – Apr 2025*

- Taught weekly Python and JavaScript programming lessons to 20 middle school students.
- Organized interdisciplinary study sessions connecting 50+ students across NSBE, Society of Women Engineers (SWE), Society of Hispanic Engineers (SHPE), and Women in Computing (WIC).

SKILLS

Programs: Python, MATLAB, C++, SolidWorks, Siemens NX, COMSOL
Skills: 3D Printing, PCB Soldering, Arduino/MPU6050 Sensor Integration